



CAPSTONE PROJECT

DEVELOPING AN AI AGENT TO SUPPORT ADMINISTRATION
AND OPERATION OF HPC SYSTEMS BASED ON SLURM

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Introduction

Motivation and Goals

Slurm powers >50% of TOP500 clusters [12] yet remains inaccessible to most users. State-changing operations affect all cluster users, and sensitive workload metadata cannot leave the site.

1. **Natural-language interface** for broad Slurm operations — inspect jobs, nodes, partitions, accounting, and submit/control workloads in plain English.
2. **Observer/Operator control flow** — read-only observation separated from state-changing operation with an explicit HITL confirmation gate.
3. **Risk-aware MCP tool integration** — typed Slurm tool interfaces with parameter validation; state-changing actions routed through confirmation.
4. **Curated benchmark** — 3,135-case dataset across 11 Slurm categories with expected tools, routing, HITL policy, and state transitions.
5. **Domain-adapted fine-tuning** — QLoRA on Qwen2.5-14B-Instruct distilled from evaluation traces; local deployment, no external API dependency.

Related Work: HPC Management & Tool-Calling Agents

System	NL	Typed Tools	HITL	Domain FT	Local	Benchmark
Open OnDemand [6]	×	×	N/A	N/A	✓	×
Chat AI [3]	×	×	×	×	✓	×
LangChain-Parsl [1]	✓	~	×	×	✓	×
Fujitsu ACB [4]	×	×	N/A	×	×	×
Gorilla / ToolLLM [9, 10]	✓	✓	×	✓	✓	General
AgentTuning [13]	✓	✓	×	✓	✓	General
τ -bench [11]	✓	✓	~	×	×	Domain
Slurm Agent (ours)	✓	✓	✓	✓	✓	✓

No existing system combines: NL interface, typed Slurm tool calls, HITL safety gate, domain-adapted fine-tuning, local deployment, and a domain-specific evaluation benchmark.

Related Work: Multi-Agent Safety & Evaluation

Work	Contribution	Gap
Greenblatt et al. [5]	AI control protocols safe under adversarial behaviour; trusted-editing + untrusted-monitoring	General; not HPC
Jamshidi et al. [7]	Tool poisoning & prompt injection risks in MCP	Threat model only; no system
AgentBench [8]	8 environments, binary pass/fail; 20–40 pp open-vs-commercial gap	No domain tools or HITL
τ -bench [11]	Policy compliance in retail/airline; GPT-4o <50%	No local model or FT
API-BLEND [2]	10k cases for API slot filling & detection	No safety or routing

None test Slurm-specific tool calling, multi-agent routing, or HITL compliance.

Our Observer/Operator instantiates Greenblatt's trusted-editing pattern; parameter validation + per-role filtering mitigates Jamshidi's MCP threats.

Research Gap

- ▶ **HPC systems** lack NL interfaces, tool-calling, and fine-tuning — users must know CLI syntax.
- ▶ **General tool-calling agents** (Gorilla, ToolLLM, AgentTuning) have no safety enforcement for destructive ops and no domain-specific evaluation.
- ▶ **Domain benchmarks** (τ -bench) provide no domain-adapted local model and no confirmation gate.
- ▶ **No existing work** combines all six: NL + typed Slurm tools + HITL + domain FT + local deploy + domain benchmark.

⇒ **This thesis fills the gap.**

Methods

System Architecture — Four-Layer Stack

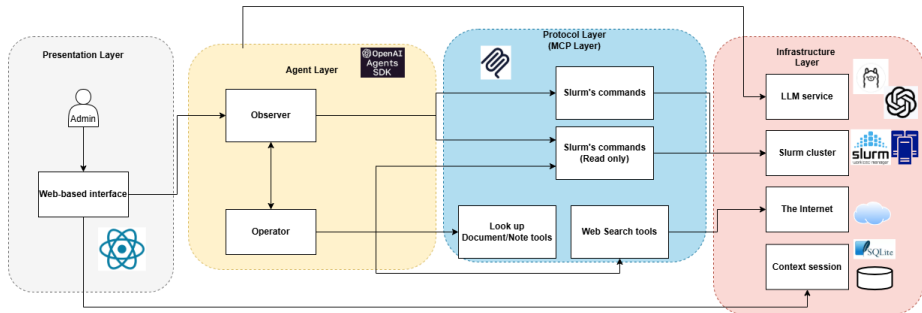
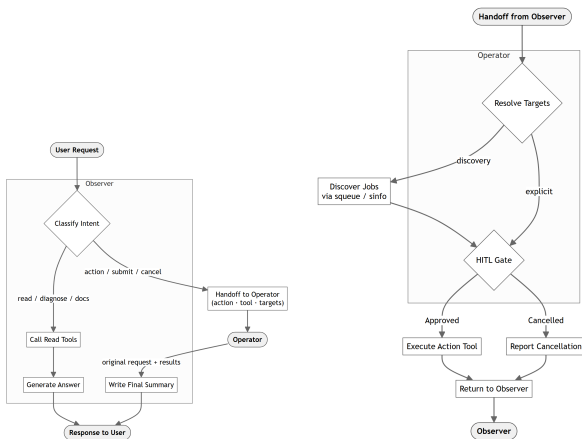


Figure: Presentation → Agent → Protocol (MCP) → Infrastructure

Observer / Operator Dual-Agent Design



Observer (read-only)

- ▶ 29 read-only tools
- ▶ Classifies intent; handles monitoring
- ▶ Issues handoff when mutation needed

Operator (state-changing)

- ▶ 40 tools (35 action + 5 read)
- ▶ Receives handoff directive
- ▶ **HITL gate** before execution

Observer *cannot* call
cancel — structural safety.

Figure: Observer routing (left) and Operator HITL confirmation (right)

Tool Partitioning: Split vs. Monolithic

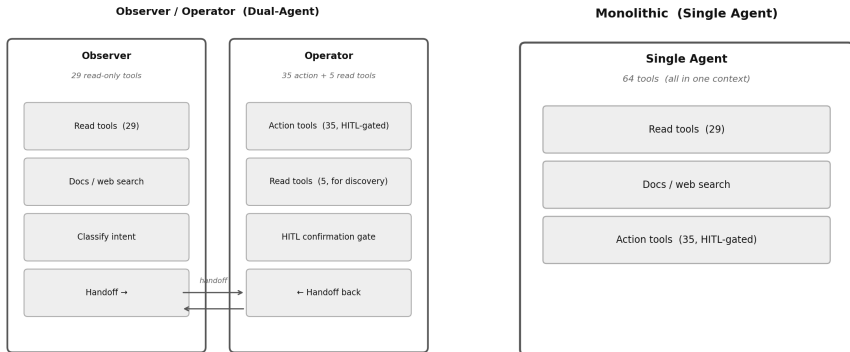


Figure: Observer/Operator split: 29 read-only + 40 tools

Figure: Monolithic baseline: all 64 tools in one context

Smaller per-agent context directly improves tool-call accuracy (+7.4 pp tool recall in ablation).

MCP Tool Layer

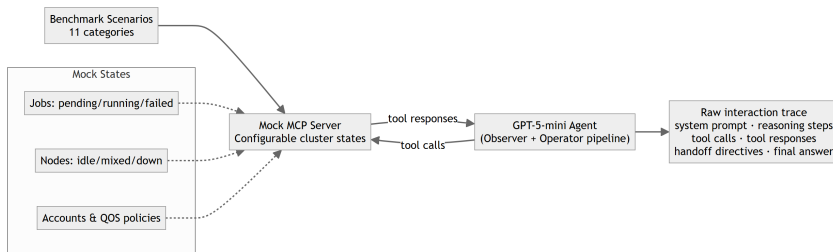


Figure: MCP tool layer: typed tools, mock engine, trace collection

- ▶ Each Slurm command as a **typed MCP tool**
- ▶ Parameter validation — no raw shell
- ▶ Structured output for model consumption
- ▶ Same layer for real Slurm & mock engine
- ▶ **64 tools:** queue, nodes, accounting, QoS, submission, control, admin, docs

Training Data Preparation

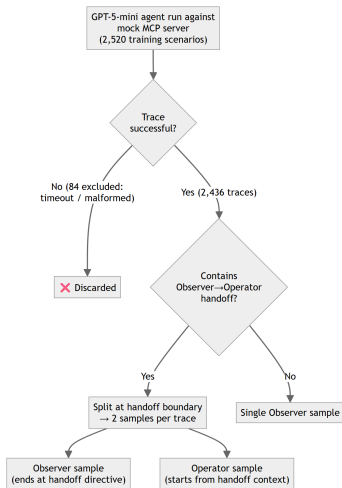


Figure: Phase 1: GPT-5-mini traces role-split at handoff boundary

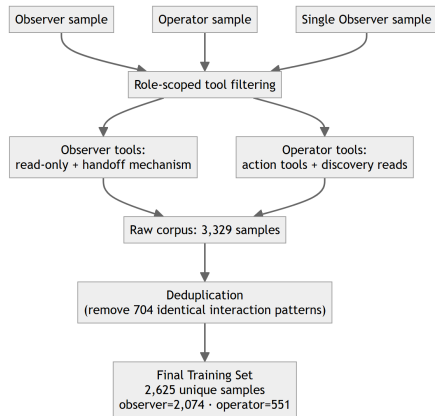


Figure: Phase 2: Role-scope filtering + dedup → 2,625 samples

Domain Adaptation: QLoRA Fine-Tuning

Model setup

- ▶ Base: Qwen2.5-14B-Instruct
- ▶ 4-bit NF4 quantization, all 48 layers
- ▶ LoRA $r=64$, $\alpha=128$ — **275M trainable params** (1.9%)
- ▶ bfloat16 adapters, 8,192 token context
- ▶ Loss on assistant tokens only

Training

- ▶ 2,625 distilled GPT-5-mini traces
- ▶ Role-scoped tool lists per sample
- ▶ 3 epochs, $lr=1e-4$, cosine schedule
- ▶ Effective batch 16 (grad accum $\times 16$)
- ▶ ~ 22 h on a single A40 (48 GB)

Experiments

Benchmark Dataset

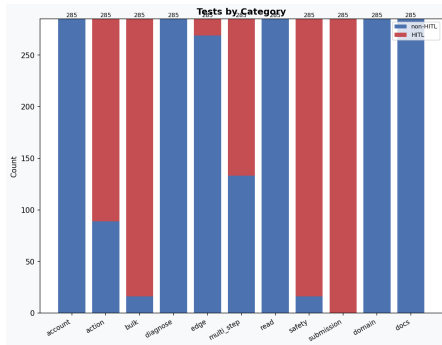


Figure: 11 categories, 285 cases each

- ▶ **11 categories:** read, diagnose, action, bulk, safety, submission, multi_step, account, edge, docs, domain
- ▶ Each case: expected tools, routing label, HITL policy, keywords, source/target state
- ▶ 80/20 stratified split, seed=42
- ▶ **615-case held-out test set** for all model comparisons
- ▶ Mock Slurm state reset before each case

Scoring Methodology

3-way comparison (dual-agent):

$$s = 0.35 r_{\text{tool}} + \mathbf{0.25} r_{\text{route}} + 0.25 r_{\text{HITL}} + 0.15 r_{\text{state}}$$

Dimension	Weight	Rationale
Tool recall	35%	Wrong tool $\Rightarrow$ wrong action
Routing match	25%	Core safety claim
HITL match	25%	Core safety claim
State match	15%	Side-effect correctness

Ablation (dual vs. mono): routing removed, weights $\times 4/3$

$$s_{\text{abl}} = \frac{7}{15} r_{\text{tool}} + \frac{1}{3} r_{\text{HITL}} + \frac{1}{5} r_{\text{state}}$$

Keyword excluded. Mean over $k=3$ trials, temperature 0.

Testing Environment

System	Arch	Adapter	Role
GPT-5-mini	Dual-agent	— (API)	Commercial baseline
FT Qwen2.5-14B-Instruct	Dual-agent	QLoRA	Ours
Base Qwen2.5-14B-Instruct	Dual-agent	None	Open-weight control
Mono Qwen2.5-14B-Instruct	Monolithic	None	Architecture ablation

Protocol

- ▶ 615-case held-out test set — identical for all models
- ▶ $k=3$ independent trials, temperature 0; mean score reported
- ▶ Mock Slurm state reset before each case
- ▶ LLM-as-judge: GPT-OSS 20B (Ollama, local), scores 0–1, reported separately

Results

Example: Bulk Job Cancellation (End-to-End Trace)

User prompt

"Cancel all of charlie's running jobs"

Observer (Think → Act → Observe)

Think: User wants to cancel multiple jobs — state-changing, must hand off.

Act: `transfer_to_operator({action: "cancel", user: "charlie", scope: "all_running"})`

Operator

Discovery read: `squeue(user="charlie")` → jobs [3001, 3002, 3007]

Pending action created — awaiting HITL...

User: "Confirm"

Execute:

`scancel(job_ids=[3001,3002,3007])` ✓

Return to Observer → final response.

Evaluation scoring for this case

Dimension	Score	Weight
Tool recall	1.00	0.35
Routing match	1.00	0.25
HITL match	1.00	0.25
State match	1.00	0.15
Overall	1.00	—

Weighted score: $s = 1.00$ ✓

Key: if agent skips HITL or uses wrong tool → lower weighted score.

Three-Way Model Comparison (615 Test Cases)

Metric	GPT-5-mini	FT Qwen (ours)	Base Qwen
Avg weighted score	95.9%	88.3%	85.2%
Tool recall	98.8%	89.2%	84.9%
Routing match	99.0%	90.1%	83.6%
HITL match	99.0%	90.4%	90.2%
State match	98.2%	90.0%	89.8%
Judge score	75.7%	79.7%	76.8%
Avg latency	28.3 s	84.8 s	80.9 s

All three systems run the Observer/Operator dual-agent architecture on the same 615-case held-out test set.

- ▶ FT Qwen closes **29% of the gap** to GPT-5-mini
- ▶ Judge score **exceeds** both base (76.8%) and commercial (75.7%)
- ▶ Local deployment: no API cost, no data transmission

Per-Category Average Scores: Three-Way Comparison



Figure: GPT-5-mini (purple) vs. FT Qwen (blue) vs. Base Qwen (orange). Largest FT gains: submission +21.7 pp, multi_step +13.8 pp, safety +11.3 pp, bulk +6.8 pp

Architecture Ablation: Dual-Agent vs. Monolithic

Metric	O/O	Mono	Δ
Avg s_{abl}	87.6%	81.6%	+6.1 pp
Tool recall	84.9%	77.5%	+7.4 pp
HITL match	90.2%	88.9%	+1.3 pp
State match	89.8%	78.9%	+10.9 pp
Judge score	76.8%	59.2%	+17.6 pp

Both scored with s_{abl} (see Scoring Methodology slide)

What the ablation shows

- ▶ +6.1 pp on same base model, same formula — pure architecture gain
- ▶ **Safety:** Observer structurally cannot call `scancel`
- ▶ **Tool scope:** +7.4 pp tool recall from halved catalog
- ▶ 378 routing-neutral cases: +10.2 pp

Ablation: Per-Category Average Score

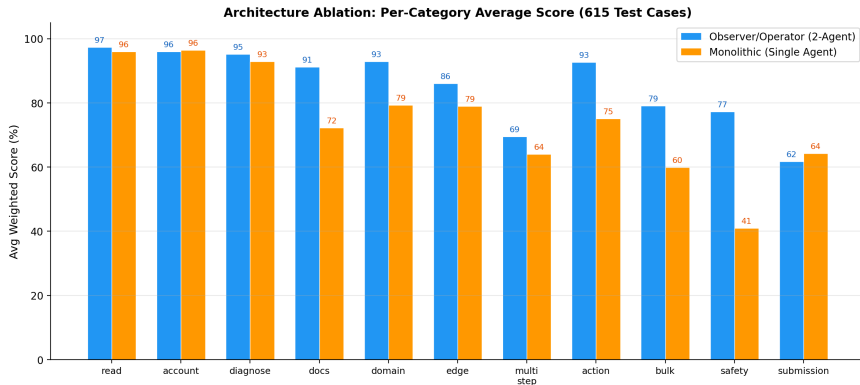


Figure: Observer/Operator vs. Monolithic per category. Action/bulk/safety categories collapse without the Operator confirmation gate.

System Demo — HITL Confirmation (Before)

What jobs are currently running?

Ready Clear

```

$ kubectl get pods --state=Running
NAME                                READY   STATUS    RESTARTS   AGE
train_model_v1-1                    1/1     Running   0           2m
inference_batch-1                    1/1     Running   0           1m

```

JOBID	NAME	USER	STATE	TIME	NODES	CPUS	MEM	PARTITION
1001	train_model_v1	alice	RUNNING	02:30:15	2	16	32G	gpu
1003	inference_batch	alice	RUNNING	01:12:08	1	4	8G	gpu

Collapse

Running jobs:

- JOBID 1001, NAME train_model_v1, USER alice, STATE RUNNING, TIME 02:30:15, NODES 2, CPUS 16, MEM 32G, PARTITION gpu
- JOBID 1003, NAME inference_batch, USER alice, STATE RUNNING, TIME 01:12:08, NODES 1, CPUS 4, MEM 8G, PARTITION gpu

Ask about your cluster... (drop files here)

1

Enter to send · Shift+Enter for newline · Drop or to attach files/images · /todo and /skill commands supported

Figure: User requests a destructive action; Observer routes to Operator for confirmation.

System Demo — HITL Confirmation (During)

What jobs are currently running?

1001 train_model_v1 alice RUNNING 02:30:15 2 16 32G gpu

1003 inference_batch alice RUNNING 01:12:08 1 4 8G gpu

Collapse

Running jobs:

- JOBID 1001, NAME train_model_v1, USER alice, STATE RUNNING, TIME 02:30:15, NODES 2, CPUS 16, MEM 32G, PARTITION gpu
- JOBID 1003, NAME inference_batch, USER alice, STATE RUNNING, TIME 01:12:08, NODES 1, CPUS 4, MEM 8G, PARTITION gpu

cancel all alice jobs

```

$ transfer to operator action_request=Cancel all jobs for user alice
required_tool=scancel
✓ ~ Handing off to Operator
$ squeue --user alice
✓ ~ 2 running

```

JOBID	NAME	USER	STATE	TIME	NODES	CPUS	MEM	PARTITION
1001	train_model_v1	alice	RUNNING	02:30:15	2	16	32G	gpu
1003	inference_batch	alice	RUNNING	01:12:08	1	4	8G	gpu

Collapse

```

$ scancel 1001,1003

```

⚠ Pending approval: scancel(job_id=1001,1003)

Please confirm or cancel.

⚠ Awaiting confirmation:

scancel(job_id=1001,1003)

✓ Confirm

✗ Cancel

Ask about your cluster... (drop files here)

Enter to send · Shift+Enter for newline · Drop or ⌘ to attach files/images · /todo and /skill commands supported

Figure: HITL confirmation gate — pending action awaiting user approval.

System Demo — HITL Confirmation (After)

What jobs are currently running?
Ready Clear

Summary of execution

Observed queue before action:

- JOBID 1001 — train_model_v1 (user: alice) — STATE: RUNNING — TIME: 02:30:15 — NODES: 2 — CPUS: 16 — MEM: 32G — PARTITION: gpu
- JOBID 1003 — inference_batch (user: alice) — STATE: RUNNING — TIME: 01:12:08 — NODES: 1 — CPUS: 4 — MEM: 8G — PARTITION: gpu

Action performed:

- cancel (scancel): Jobs 1001, 1003 cancelled successfully.

are they still running?

```

$ squeue --job_id 1001,1003
$ No such job: 1001,1003.

No such job: 1001,1003.

$ sacct --user alice --starttime now-1day --format
JobID,JobName,User,State,ExitCode,Elapsed
$ JobID|JobName|User|State|ExitCode|Elapsed

JobID|JobName|User|State|ExitCode|Elapsed
-----
1001|train_model_v1|alice|CANCELLED|0:0|02:30:15|16|32G
1003|inference_batch|alice|CANCELLED|0:0|01:12:08|4|8G

```

[Collapse](#)

They are not running.

Historical records (sacct, last 1 day) for alice:

- JOBID 1001, NAME train_model_v1, STATE CANCELLED, EXITCODE 0:0, ELAPSED 02:30:15
- JOBID 1003, NAME inference_batch, STATE CANCELLED, EXITCODE 0:0, ELAPSED 01:12:08

Ask about your cluster... (drop files here)

Enter to send · Shift+Enter for newline · Drop or to attach files/images · /todo and /skill commands supported

Figure: After approval: confirmed execution with result summary.

System Demo — RAG Documentation (Before)



Figure: User asks a Slurm documentation question.

System Demo — RAG Documentation (After)

I'm new to this cluster, can you explain wha...

Ready

Clear

3. Match access and accounting

- Some partitions are restricted to certain groups/accounts or QOS levels. Use a partition you are permitted to run in (your account or project may restrict choices).

4. Choose for turnaround vs throughput

- Short/debug partitions (short walltime, small node-count) are faster for tests. Long/production partitions are for full runs but can have longer queue waits or lower priority.

5. Consider scheduling policies

- Some partitions may be preemptible or have reservations; check if preemption or reservations matter for your run.

6. Software / Feature availability

- Some partitions map to nodes with specific software, network, or storage attachments. Pick one that has the libraries or filesystem performance you need.

Commands to inspect your cluster (what to run)

- sinfo

 - shows partitions, node counts and general node capabilities.
- squeue -p <partition>

 - shows current queue depth and waiters for that partition.
- scontrol show partition=<name>

 - shows partition configuration (limits, allowed accounts/QOS, flags).

Practical quick workflow

- sinfo — see which partitions exist and which advertise GPUs / highmem.
- squeue -p <candidate_partition> — check how busy it is.
- scontrol show partition=<name> — check time limits, max nodes, and access.
- Submit a small test to a short/debug partition first, then move to a production partition for the full run.

If you want, tell me the resources your job needs (CPUs, memory, GPUs, expected runtime, account/project) and I can show which partitions on this cluster look appropriate (I'll run sinfo/squeue for you).

Ask about your cluster... (drop files here)

↑

Enter to send · Shift+Enter for newline · Drop or to attach files/images · /odo and /kill commands supported

Figure: Agent retrieves and synthesizes relevant documentation via RAG tools.

System Demo — Web Search (Before)

My CUDA job crashed with an out of memory er...

Ready Clear

My CUDA job crashed with an out of memory error, can you help me quick search online to understand what happened and how to fix it?

```

$ web_search --search_type error CUDA out of memory error causes and fixes
↳ [WEB_SEARCH]:

[WEB_SEARCH]:
Query: CUDA out of memory error causes and fixes
Search type: error
Results:
1. Out-of-memory errors running pbrun fq2bam through singularity on A100s ... - Hello,
URL: https://forums.developer.nvidia.com/t/out-of-memory-errors-running-pbrun-fq2bam-f
2. OOM-killer on slurm based cluster | by Mayin Dev | Medium - Out-of-memory (OOM) er
URL: https://medium.com/@mayintuji/oom-killer-on-slurm-based-cluster-93e5865811d6
3. CUDA out of memory even after using DistributedDataParallel - I try to train a big
URL: https://discuss.pytorch.org/t/cuda-out-of-memory-even-after-using-distributeddata
Tip: call fetch_web_content(url=...) to retrieve full page text.
⏪ ⏩

⏶ Collapse
$ fetch_web_content https://forums.developer.nvidia.com/t/out-of-memory-errors-running-
pbrun-fq2bam-through-singularity-on-a100s-via-slurm/239972
↳ [WEB_FETCH]:

[WEB_FETCH]:
URL: https://forums.developer.nvidia.com/t/out-of-memory-errors-running-pbrun-fq2bam-f
⏪ ⏩

⏶ Show all (198 lines)
$ web_search --search_type error torch.cuda.OutOfMemoryError fixes reduce batch size
gradient checkpointing torch.cuda.empty_cache
↳ [WEB_SEARCH]:

[WEB_SEARCH]:
Query: torch.cuda.OutOfMemoryError fixes reduce batch size gradient checkpointing torc
⏪ ⏩

⏶ Show all (11 lines)

```

Ask about your cluster... (drop files here)

Enter to send · Shift+Enter for newline · Drop or 📎 to attach files/images · /todo and /skill commands supported

↑

Figure: User asks a question requiring external knowledge.

System Demo — Web Search (After)

My CUDA job crashed with an out of memory er... Ready Clear

- `torch.cuda.max_memory_allocated()`, `torch.cuda.memory_reserved()`
- Add checkpoints in your training loop to see where peak memory rises.
- If memory grows every epoch/step, suspect a leak (unintended accumulation of tensors, missing `.detach()`).

Cluster/container-specific checks and fixes

- Make sure your Slurm script requests the GPUs (example):
 - `#SBATCH--gres=gpu:1` (or `gpu:a100:4`, etc.)
- If using Singularity/Apptainer: run a short interactive job on the target node and inside container run `nvidia-smi` to confirm GPUs are visible.
- If other jobs are on the same node, they can occupy GPU memory; `nvidia-smi` shows processes. If so, run on an idle node or request exclusive access if available (consult your cluster admin/policies).
- On multi-GPU training: ensure your code uses the correct device mapping for `DistributedDataParallel` or `DataParallel`; incorrect mapping can overload one GPU.

If OOM persists after the above

- Try a smaller model or a trimmed input to confirm capacity limits.
- Reproduce locally on a single GPU (if possible) to iterate quickly.
- Consider model parallelism or distributed training to spread memory across GPUs.
- For TensorFlow, enable per-process memory growth (`tf.config.experimental.set_memory_growth`).

Common pitfalls that masquerade as OOM

- Container not exposing GPUs to the job (looks like OOM because device access fails).
- `CUDA_VISIBLE_DEVICES` mismatch: your code uses GPU 0 but Slurm assigned a different physical GPU.
- Memory fragmentation from many allocations: restarting the process can temporarily fix it.

If you want, provide:

- the exact error traceback (copy/paste),
- the key parts of your Slurm script (gres, partition, any container flags),
- and whether you run inside a container, and I can give targeted fixes and a short checklist of commands to run on the node (`nvidia-smi`, torch memory calls) to pinpoint the cause.

Ask about your cluster... (drop files here) ↑

Enter to send · Shift+Enter for newline · Drop or  to attach files/images · /docs and /kill commands supported

Figure: Agent performs web search and returns a cited answer.

Conclusion

Summary of Contributions

1. **Broad Slurm intent surface** — 11 categories, 64 typed MCP tools covering monitoring through confirmed destructive operations
2. **Observer/Operator workflow** — structural read/write separation with pending-action HITL confirmation; safety by construction, not prompt compliance
3. **3,135-case benchmark dataset** — manually constructed, scored on tool recall, routing, HITL, state, and judge quality
4. **Scenario-grid evaluation system** — architecture-level metrics, parallel workers, persistent result artifacts, inspectable traces
5. **Domain-adapted fine-tuning** — Qwen2.5-14B-Instruct via QLoRA distilled from commercial traces; **88.3% mean weighted score**, closes 29% of the gap to GPT-5-mini, fully local

Limitations & Future Work

Limitations

- ▶ Synthetic benchmark only — no live-cluster user study
- ▶ Routing metric (25%) structurally disadvantages monolithic baseline
- ▶ Training data from finite mock state space
- ▶ Single-turn evaluation only
- ▶ No multi-tenant auth / audit logging

Future Work

- ▶ Real-cluster validation with live scheduler variability
- ▶ Multi-turn conversation benchmark
- ▶ Role-aware authentication & audit logging
- ▶ Larger base model (70B) for harder categories
- ▶ Practitioner user studies

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Thank you for listening!

Appendix

Fine-Tuning Hyperparameters

Parameter	Value
Base model	Qwen2.5-14B-Instruct
Quantization	NF4 + double quantization
LoRA rank / α	64 / 128
Target layers	All 48 layers
Target modules	q/k/v/o_proj, gate/up/down_proj
Trainable parameters	~275M (1.9%)
Training samples	2,625 unique (3,329 raw – 704 deduped) (2,074 Observer + 551 Operator)
Teacher model	GPT-5-mini (distilled traces)
Epochs	3
Effective batch size	16 (micro=1, grad_accum=16)
Max sequence length	8,192 tokens
Learning rate	1×10^{-4}
LR schedule	Cosine with 5% warmup
Hardware	1× NVIDIA A40 (46 GB)

Training Loss Curve (WandB)

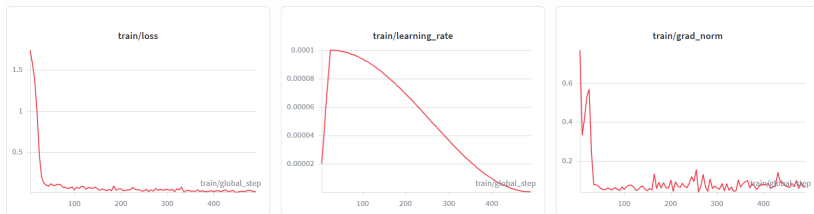


Figure: Training loss over 3 epochs on 2,625 distilled samples. Cosine LR schedule with 5% warmup; loss on assistant tokens only.

Benchmark Scenario Distribution

Dataset Distribution by Scenario

